

Type 1 & Type 2 Errors

Confusion Matrix

	1	0	Actual
1	TP	FP	→ Type 1 Error
0	FN	TN	

Predicted ↓
Type 2 Error

Reality - Null hypothesis is True OR
Null hypothesis is False

Decision - Null hypothesis is True OR
Null hypothesis is False

Outcome 1 - We reject the null hypothesis
→ When in reality it is False

Outcome 2 - We Fail to reject the null hypothesis OR
we retain the null hypothesis

Outcome 3 - We reject the null hypothesis
when in reality it is True
Type 1 error

Outcome 4 - We Fail to reject the null hypothesis
In reality it is False
Type 2 error

Type 1 error (False Positive, α)

Definition - You conclude there is a difference or effect, but there is not

Example - A pregnancy test says a male user is pregnant; the test wrongly rejected the true null hypothesis (i.e. not pregnant)

Probability - The risk is denoted by alpha (α), often set at 0.05 (5%)

Type 2 Error (False Negative, B)

Definition - You Fail to reject a null hypothesis that is actually False, missing a real effect

Example - A cancer test tells a patient they are cancer-free, but they actually have the disease; it failed to reject the

null hypothesis (no cancer)

Probability - The risk is denoted by β

Relationship

Type 1 and Type 2 errors are inversely related; reducing the risk of one usually increases the risk of the other.

H_0 refers to Null hypothesis

Error Type 🔗	Definition	Alternative Name	Reality	Decision
Type 1	Reject H_0 when true	False Positive	H_0 True	Reject H_0
Type 2	Fail to reject H_0 when false	False Negative	H_0 False	Fail to reject H_0